

SUTTSU, Japan

WMO: 474210

Lat: 42.7956N

Lon: 140.2244E

Elev: 35

StdP: 100.90

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.6	(c) -7.4	(d) -14.5	(e) 1.1	(f) -6.9	(g) -13.3	(h) 1.2	(i) -5.8	(j) 11.3	(k) -0.9	(l) 10.3	(m) -0.9	(n) 4.9	(o) 310	(p) 0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.4	(c) 27.1	(d) 22.9	(e) 25.6	(f) 21.9	(g) 24.4	(h) 21.3	(i) 23.6	(j) 26.1	(k) 22.8	(l) 25.0	(m) 22.0	(n) 23.8	(o) 3.4	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.6	25.2	22.0	16.7	24.4	21.2	15.9	23.6	71.1	26.2	67.9	25.1	64.7	23.9	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 8.9	(c) 7.9	(e) -10.4	(f) 30.0	(g) 1.6	(h) 1.8	(i) -11.5	(j) 31.3	(k) -12.5	(l) 32.3	(m) -13.4	(n) 33.3	(o) -14.6	(p) 34.6		
(a) 10.2	(b) 8.9	(c) 7.9	(e) -11.2	(f) 24.9	(g) 1.6	(h) 1.2	(i) -12.3	(j) 25.8	(k) -13.3	(l) 26.5	(m) -14.1	(n) 27.2	(o) -15.3	(p) 28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-2.6	-2.1	1.1	6.5	11.6	15.5	19.8	21.5	18.2	12.0	5.3	-0.4
	DBStd	8.88	2.66	2.85	2.90	3.18	2.98	2.60	2.49	2.28	2.60	3.03	3.90	3.10
	HDD10.0	1620	389	339	277	115	16	1	0	0	0	15	146	321
	HDD18.3	3623	648	573	536	356	208	90	12	2	34	195	390	579
	CDD10.0	1234	0	0	0	9	67	167	304	358	246	78	6	0
	CDD18.3	196	0	0	0	0	1	6	59	101	29	0	0	0
	CDH23.3	537	0	0	0	1	6	16	144	320	51	0	0	0
	CDH26.7	61	0	0	0	0	0	1	18	39	2	0	0	0
Wind	WSAvg	3.7	4.1	4.1	3.9	3.9	4.0	3.6	3.4	3.2	3.1	3.3	3.7	4.1
Precipitation	PrecAvg	1265	124	90	68	60	67	60	97	129	150	130	149	140
	PrecMax	1678	198	173	213	118	151	135	248	286	325	229	257	208
	PrecMin	943	58	52	18	12	13	7	23	41	56	59	77	75
	PrecStd	179	33	29	39	30	32	36	57	63	73	48	49	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	6.4	11.2	19.4	23.8	25.4	28.9	29.8	26.7	20.8	16.1	9.9
		MCWB	2.9	3.8	6.5	11.7	15.1	18.6	23.6	24.4	22.7	16.3	12.2	7.7
	2%	DB	3.4	4.5	8.4	15.8	20.6	22.9	26.5	27.6	24.7	19.1	14.2	7.4
		MCWB	1.6	2.2	4.9	10.0	13.8	17.6	22.3	23.5	21.1	15.5	11.3	5.4
	5%	DB	2.3	3.2	6.8	13.5	18.3	21.2	24.9	26.2	23.4	17.8	12.7	5.6
		MCWB	0.5	1.1	3.9	8.4	12.5	17.2	21.7	22.6	20.3	14.7	10.1	3.5
	10%	DB	1.3	2.1	5.4	11.5	16.5	19.7	23.5	25.0	22.3	16.7	11.3	4.1
		MCWB	-0.6	0.2	2.9	7.5	11.7	16.4	20.8	22.0	19.4	13.9	8.8	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	4.1	7.1	12.9	16.4	19.9	24.2	25.2	23.5	17.6	13.4	8.2
		MCDB	4.3	5.9	10.0	17.5	21.9	23.3	27.2	28.5	25.6	19.1	15.3	9.5
	2%	WB	2.0	2.6	5.4	10.6	14.8	18.7	23.1	24.1	22.3	16.3	11.8	5.6
		MCDB	3.2	4.1	7.7	14.7	19.1	21.4	25.7	26.6	23.9	18.1	13.6	7.2
	5%	WB	0.8	1.4	4.3	9.1	13.6	17.7	22.2	23.3	21.1	15.4	10.4	3.7
		MCDB	2.1	2.9	6.5	12.4	16.9	20.2	24.2	25.5	22.7	17.3	12.4	5.3
	10%	WB	-0.4	0.3	3.1	7.9	12.6	16.8	21.3	22.5	19.9	14.3	9.0	2.2
		MCDB	1.1	2.0	5.1	10.9	15.5	19.3	23.1	24.5	21.8	16.4	11.1	3.9
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.2	4.9	6.5	6.9	6.1	5.4	5.4	6.1	6.3	5.3	4.1
		MCDBR	5.2	5.9	7.4	10.1	10.6	8.7	7.4	6.8	6.8	7.2	7.5	6.1
		MCWBR	4.7	5.2	5.5	6.0	5.8	4.9	4.0	3.8	4.2	4.9	6.1	5.6
	5% WB	MCDBR	5.3	5.8	7.0	8.8	9.1	7.5	6.3	6.0	5.5	6.3	7.3	6.1
		MCWBR	5.2	5.4	5.7	5.7	5.4	4.5	3.6	3.6	4.0	5.0	6.5	5.9
Clear-Sky Solar Irradiance	taub	0.359	0.409	0.476	0.535	0.552	0.527	0.512	0.474	0.420	0.414	0.400	0.378	
	taud	2.049	1.913	1.828	1.777	1.827	1.973	2.088	2.213	2.311	2.170	2.104	2.031	
	Ebn at Noon	730	750	748	740	742	763	771	789	807	748	679	660	
	Edh at Noon	111	150	186	211	206	179	158	135	114	115	105	104	
All-Sky Solar Radiation	RadAvg	1.15	1.88	3.25	4.51	5.03	4.83	4.30	4.32	3.76	2.68	1.47	0.97	
	RadStd	0.09	0.20	0.28	0.46	0.49	0.50	0.45	0.35	0.39	0.16	0.16	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page